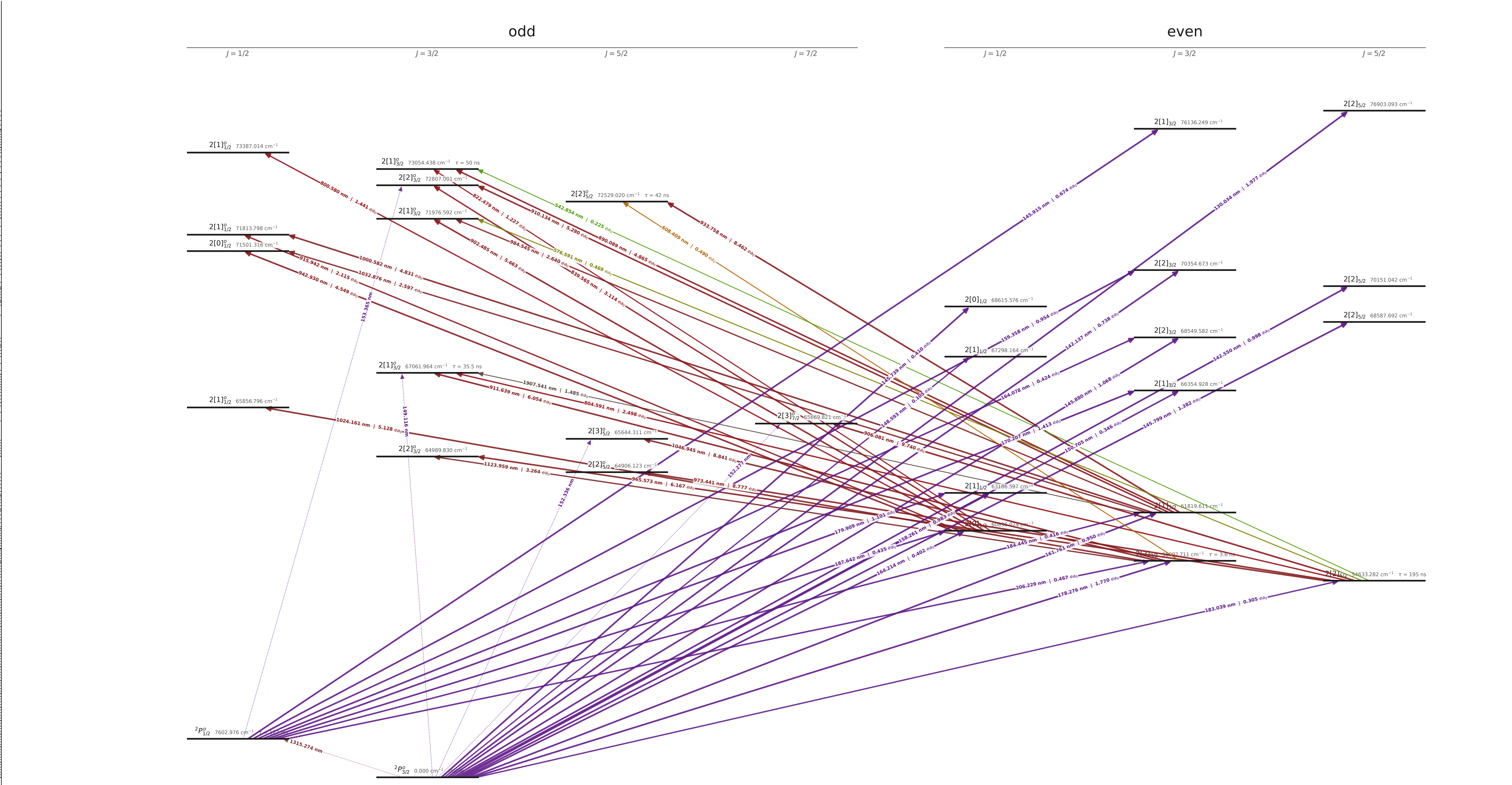


Iodine

I (I I). Strongest classified transitions below 2 μm (NIST): 29 levels, 50 lines.

I⁺ ionisation limit 84294.90 cm⁻¹ = 10.45124 eV (118.631 nm)

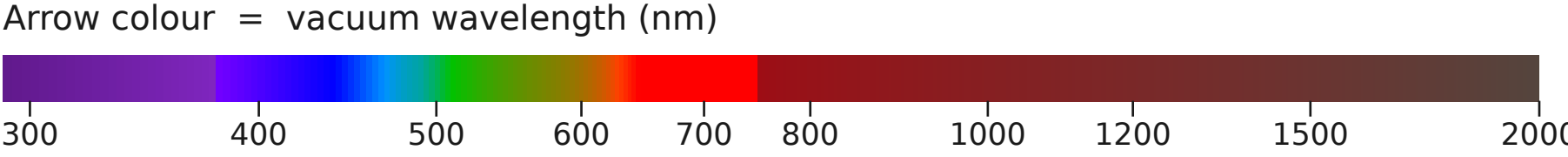


Key transitions

Transition	λ vac (nm)	λ air (nm)	ν (THz)	⟨J er J'⟩ (ea ₀)	A (s ⁻¹)	Γ/2π up. (MHz)	branch (%)	Use
² P _{3/2} ^o – 2[1] _{5/2} ^o	149.1158	–	2010.4670(3)	–	–	4.483	–	TALIF detection of ground-state I: 2 x 298.2 nm, fluoresc...
² P _{3/2} ^o – 2[3] _{7/2} ^o	152.2770	–	1968.7316(3)	–	–	–	–	TALIF of ground-state I near 2 x 304.5 nm, fluorescence o...
² P _{3/2} ^o – 2[3] _{5/2} ^o	152.3361	–	1967.9669	–	–	–	–	classic TP-LIF /
² P _{3/2} ^o – 2[2] _{5/2} ^o	153.3648	–	1954.7675	–	–	–	–	TP-LIF detection of spin-orbit excited I*
² P _{3/2} ^o – 2[2] _{3/2} ^o	178.2763	–	1681.6172	1.770	2.800e+08	44.21	100	178.28 nm resonance line
² P _{3/2} ^o – 2[2] _{1/2} ^o	183.0386	–	1637.8646	0.305	5.140e+06	0.8162	100	183.04 nm resonance line
2[2] _{5/2} – 2[1] _{5/2}	804.5905	804.3693	372.6025	2.498	6.070e+06	4.483	21.6	804 nm fluorescence line used to detect I atoms after two...
2[2] _{5/2} – 2[3] _{7/2}	906.0811	905.8325	330.8671	9.740	3.230e+07	–	–	905 nm fluorescence line: the only decay channel of 6p 2[...
2[2] _{3/2} – 2[1] _{5/2}	911.6391	911.3890	328.8499	6.054	2.450e+07	4.483	87	911 nm fluorescence channel of the TALIF upper level 6p 2...
² P _{3/2} ^o – ² P _{1/2} ^o	1315.2744	1314.9147	227.931501(15)	–	8.000e+00	1.253e-06	100	1315 nm iodine-laser line

λ, ν from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST.
Γ/2π: natural linewidth of the upper level, 1/(2πτ).

Arrow label: vacuum wavelength | reduced dipole matrix element |⟨J||er||J'⟩| in ea₀
Matrix elements are measured or NIST values; * = high-accuracy theory, ≈ = model or semi-empirical calculation.
Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, n = 70.
Level caption: energy (NIST, or a measurement of the last 20 years where one exists) and measured lifetime (where one exists).



¹²⁷I hyperfine structure (I = 5/2)

Level	A (MHz)	B (MHz)	Shift of each F level from the centre of gravity (MHz)
² P _{3/2}	827.265(3)	1146.356(10)	F=1: -3540.69 F=2: -2803.25 F=3: -837.31 F=4: +3388.83
² P _{1/2}	6587(4)	–	F=2: -11527.25 F=3: +8233.75
2[2] _{5/2}	1417.100	-1065.500	F=0: -13145.48 F=1: -11472.66 F=2: -8222.91 F=3: -3588.03 F=4: +2144.30 F=5: +8590.50
2[3] _{7/2}	586.700	-1104.400	F=1: -7192.02 F=2: -5703.07 F=3: -3587.99 F=4: -988.75 F=5: +1905.30 F=6: +4857.52

Measured A, B constants; sources in the data file.